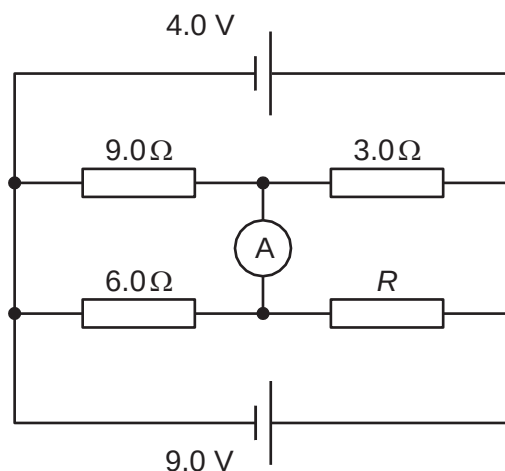


- 23** In the circuit shown, the cells have negligible internal resistance and the reading on the ammeter is zero.



What is the resistance of R ?

- A** $2.0\ \Omega$ **B** $6.0\ \Omega$ **C** $12\ \Omega$ **D** $18\ \Omega$

Answer: C

The ammeter reads zero which means the potentials at the two ends of the ammeter are the same. This means that the potential difference across the $9.0\ \Omega$ resistor is the same as that across the $6.0\ \Omega$ resistor.

$$\frac{V_{9.0\Omega}}{4.0} = \frac{9.0}{9.0 + 3.0}$$

$$V_{9.0\Omega} = 3.0\text{ V}$$

$$\frac{V_{6.0\Omega}}{9.0} = \frac{6.0}{6.0 + R}$$

$$\frac{3.0}{9.0} = \frac{6.0}{6.0 + R}$$

$$V_{9.0\Omega} = 3.0\text{ V}$$

$$R = 12\ \Omega$$

24 Electrons, each of mass m and charge e , are accelerated from rest in a vacuum through a